DATA ANALYTICS — STAGE 0 TASK

Stage 0 (Data Analysis) First Insights & Visual Thinking

Requirements / Task Brief

This task introduces you to the data analytics workflow with a focus on early exploration, visual thinking, and communication. You will work with a real dataset, explore it quickly, create simple visuals, and present your first insights clearly and professionally. It is about how you approach data and communicate what you see.

Datasets : Datasets Folder

Evaluation Criteria / Acceptance Criteria

Step 1: Select a Dataset

You will be provided with five (5) curated datasets to choose from.

- Select ONE dataset only
- Do not use any dataset outside the provided list

Step 2: Explore the Dataset

Before deep analysis:

- Understand the dataset structure
- At this stage, focus on what you notice, not perfection

Step 3: Identify Your Curiosity

Define three (3) questions that guide your exploration.

Your questions should be:

- Answerable with the dataset
- Interesting and relevant
- Visual-friendly (can be explored using charts)

Step 4: Create Charts and Reports

You will create charts and supporting analysis to explore your questions.

Tool Requirement (Mandatory)

- **Excel only**

Chart and Report Guidelines

- Create 4–6 charts that answer your questions
- Focus on clarity and insight, not aesthetics
- No complex calculations required
- One dataset, one story

Step 5: Document Your Work

Prepare a short report that explains:

- What the dataset is about
- What you noticed at first glance
- What questions you explored
- What stood out after visualizing the data

Step 6: Identify Limitations

Briefly acknowledge any limitations, such as:

- Limited time period
- Assumptions made during exploration

Submission Format

Report Format

- **6-slide presentation**
- Must be submitted in **PDF format**
- **Content must be clean, readable, and professional**
- **Your story should flow logically: Dataset → Observations → Questions →Visuals → Insights**

Excel File

- **Include link to your Excel file containing all charts and analysis**

- **Ensure link is accessible (view-only)**

Social Media Post (Mandatory)

Share a 1-minute video presentation of your analysis:

- **Post on LinkedIn or Twitter/X**
- **Tag @HNG**
- **Use #HNGDataAnalytics**

Deliverables

- **Link to the PDF report (3–5 pages) OR PDF presentation (6 slides)**
- Link to Excel file with charts
- Social media post link **(1-minute video presentation)**
- Submit the link to your post in the submission form

Submission Link

https://docs.google.com/forms/d/e/1FAIpQLScm4xEv4SoTPp5YxNPQXnKYfODbp_mEASTRZXXhHV4J7nKrTA/viewform

Points

10

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Pass Mark

6

Format: Integer

DATA ANALYTICS — STAGE 1 TASK

Stage 1 Task: Data Cleaning, Preparation, and Title Optimization

Requirements / Task Brief

This task focuses on ensuring the dataset is clean, reliable, and ready for further marketing analysis while introducing a new short_title feature for SEO-optimized and concise product titles. You will explore, identify, and resolve common data issues using Excel. Additionally, you'll implement logic to generate shorter product titles for each entry, ensuring they retain key information while being concise and prepare a detailed technical report.

Product Data : **Dataset Link**

Evaluation Criteria / Acceptance Criteria

-1- Dataset Familiarization

Download and review the dataset to understand its structure and key variables.

- Identify significant variables for cleaning and for creating the short_title feature.
- Check column names for clarity and consistency.

-2- Initial Data Exploration

- Conduct a high-level review of the dataset using descriptive statistics.
- Identify data quality issues such as missing values, duplicates, and inconsistent formats.
- Use Excel for initial exploration.

-3- Data Cleaning Process

- **Handling Missing Values:**Identify and evaluate columns with missing data.
- Apply appropriate strategies.
- **Removing Duplicate Entries:**Locate and remove duplicate records to ensure data accuracy.
- **Standardizing Data Formats:**Ensure uniform formats for dates, currencies, and other fields.
- Standardize column names (e.g., sales_amount instead of SalesAmount).
- **Verifying Data Accuracy:**Check critical variables for anomalies (e.g., negative prices or invalid ratings).

-4- Creating the short_title Feature

- **Objective:**Generate a concise version of product titles that retains essential information for SEO and readability.
- **Steps:**Analyze original product titles to identify key components (e.g.,product name, category, attributes).
- Remove redundant phrases or words (e.g.,"includes," "set of," "features").
- Limit titles to 30–50 characters, focusing on essential details and keywords.

Implementation Example:

- **Original Title:** "Tulip Flowers Blackout Curtain for Door, Window & Room | Eyelets &Tie Back | Canvas Fabric | Set of 2 PCS" **Short Title:** "Tulip Blackout Curtain - 2PCS"
- **Original Title:** "Marks & Spencer Girls' Pyjama Sets T86_2561C_Navy Mix_9-10Y" **Short Title:** "Girls' Navy Pyjama Set - 9-10Y"

-5- Documentation and Reporting

Prepare a professional report documenting the data cleaning and title optimization process:

- **Introduction:** Briefly describe the dataset and task objectives.
- **Data Cleaning:** Summarize issues identified and the cleaning steps taken.
- **Short Title Creation:** Explain the methodology and examples of optimized titles.
- **Clean Dataset Overview:** Highlight key statistics and improvements in the dataset.
- **Include screenshots or visualizations illustrating the impact of data cleaning and title optimization.**

-6- Tool Requirement

- Make use of **EXCEL ONLY** as the Data Analysis tool for the data cleaning, preparation and optimization task.

Submission Format

Deliverables

Cleaned Dataset

- Updated dataset (no missing values, duplicates, inconsistencies)
- Must include the new short_title column
- Upload as a file (Excel) via Google Drive link.

Technical Report (PDF)

Your report should include:

- Introduction & objectives
- Data cleaning steps
- Summary of improvements, with examples of original and short titles.

NOTE!

Your links should be google drive links. Change default setting on each google drive file to enable anyone to have access to view.

Submission Link

https://docs.google.com/forms/d/e/1FAIpQLSfXgRZSyJ-G0SrSovmwvim-lo1DNxoJaQGCOLHC8FDGY-PCXg/viewform

Points

10

Format: Integer

Pass Mark

7

Format: Integer

Format: Integer

DATA ANALYTICS — STAGE 2 TASK

Hello, Data Analysts!
Welcome to Stage 2: SQL with Business Context.
Congratulations on making it this far. Now it gets deeper.

TASK EXPLAINER AIRTABLE

This stage isn't just about writing SQL queries — it's about thinking like a business-focused data analyst.

Business Scenario
You have joined the data team at **TradeZone**, a fast-growing Nigerian e-commerce platform connecting buyers and sellers across Lagos, Abuja, Kano, Port Harcourt and Ibadan. Over the past two years (2023–2024), the platform has grown quickly but that growth is masking some serious operational problems. Customer retention is dropping. Some sellers are dragging down overall platform ratings. Certain product categories are underperforming despite heavy promotion spend.

The Head of Growth and the Head of Seller Operations have requested a combined data review ahead of the 2025 planning cycle. They do not want a slide deck of charts. They want to understand what is actually happening in the business and what to do about it.

Your job is to clean the data, answer eight specific business questions using SQL and deliver a concise analyst memo that turns your query results into decisions the leadership team can act on.

The Dataset
The TradeZone database

Tool Requirement
Postgres

Part A: Data Cleaning & Preparation
Before writing any analysis queries, you must clean the dataset. Your SQL file must contain a clearly labelled Cleaning section that addresses each of the following issues. Document any decision you make if you drop a record, explain why in a SQL comment.

- Missing Values: Identify and handle NULL or blank entries in critical columns
- Duplicate Records: Check for and remove duplicate rows in customers, sellers and orders.
- Inconsistent Formatting: Standardise city names. Ensure all date columns follow a consistent format (YYYY-MM-DD). Normalise product category names to title case.
- Data Validation: Verify that each order's total_amount matches the sum of its line items in order_items. Flag orders where the difference is greater than ₦10. Check that all review ratings are between 1 and 5. Check for negative product prices or discount percentages above 100%.

Part B: Business Questions SQL
Using your cleaned data, write SQL to answer each of the following eight questions. Number each query clearly to match the question. Every query must be commented on with the business question it answers.

Question 1: Customer Acquisition & 30-Day Conversion
Find the top 5 states by number of new customer sign-ups in 2024. For each state, calculate what percentage of these new customers made at least one purchase within their first 30 days of signing up.

Question 2: Product Performance
Identify the top 10 products by total revenue in 2024. Include product name, category, total revenue and total number of orders. Sort by revenue descending.

Question 3: Seller Fulfilment Efficiency
Calculate the average time in hours between order placement and delivery for each seller. Return the top 20 sellers with the fastest average fulfilment times among sellers who have completed at least 20 orders. Include their total completed orders and average customer rating.

Question 4: Quarterly Revenue Trends
Compare quarterly revenue across 2023 and 2024. For each quarter, calculate total revenue, average order value and total number of orders. Identify which single quarter showed the strongest revenue growth from 2023 to 2024.

Question 5: Customer Spend Segmentation
Segment customers based on their total spend in 2024 into three groups:

- High Spenders: ≥ ₦100,000
- Medium Spenders: ₦50,000 – ₦99,999
- Low Spenders: < ₦50,000

For each group, calculate the customer count, average spend per customer and total revenue contribution.

Question 6: Payment Method Preferences by State
Analyse payment method preferences across each state in the dataset. For each state, show the transaction count and total amount for each payment method (Cash on Delivery, Card, Mobile Money, Bank Transfer) and identify the most popular method per state.

Question 7: Review Ratings and Sales Performance
Group products based on their average review rating into three categories:

- High Rated: 4.0 and above
- Mid Rated: 3.0 – 3.99
- Low Rated: Below 3.0

For each category, calculate the product count, total revenue and average unit price.

Question 8: Top Seller Bonus Qualification
Identify the top 10 sellers in 2024 by total revenue who completed at least 10 orders and have an average customer rating of 4.0 or above. Include their total orders, average rating, and total revenue.

Part C: The Analyst Memo
This is the most important part of Stage 2. Your SQL queries are the evidence. The memo is what the business actually uses. Write a structured memo (minimum 500 words) addressed to the Head of Growth and Head of Seller Operations.

Required Structure
1. Executive Summary
Two to three sentences only. What is the single most important thing leadership needs to know about TradeZone's performance across 2023–2024?

2. Key Findings Three Maximum
Select the three most significant findings from your eight queries. For each finding: state the specific number or trend, explain what it means for the business and name which query it came from.

3. Recommendations Two Required
Make exactly two recommendations based on your findings. Each must state: the specific action to take, the team or person responsible and the outcome you would expect within 60 to 90 days.
Recommendations must be traceable to a query result and do not recommend things your data does not support.

4. Data Quality Notes
Briefly describe the two most significant data quality issues you found and what decisions you made. If those decisions turned out to be wrong, what would the impact be on your analysis?

5. What the Data Cannot Tell Us
Name one business question the leadership team might ask that your current dataset cannot answer and state what additional data you would request to answer it.

Deliverables
Ensure your Google Drive folder is named exactly **HNG14-DA-S2** and is set to **"Anyone with the link can view"** before submitting.
Your folder must contain the following files, named exactly as specified:

1. **Data Cleaning Script** — Part 1.sql *(your Part A data cleaning work)*
2. **SQL Query Files** — Q1.sql, Q2.sql, Q3.sql, Q4.sql, Q5.sql, Q6.sql, Q7.sql, Q8.sql *(one file per query)*
3. **Database Dump** — cleaned_dump *(saved with a* .sql*,* .dump*, or* .txt *extension)*
4. **Analytical Report** — Analyst_Memo.pdf *(your Part C memo exported as a PDF)*

NOTE: File and folder names must match exactly as written above — including capitalization and spacing — to ensure your submission is graded correctly by the automated system.

Submission:
There is no Google Form for this stage. To submit, follow the steps below:

1. Go to the **stage-2-da** Slack channel
2. Type and send the **/submit** command
3. A **submission modal/form will pop up** — fill in the following:

- **Google Drive Folder Link** — the link to your HNG14-DA-S2 folder containing all required files
- **PDF Report Link** — the direct link to your Analyst_Memo.pdf file within the folder

Deadline: April 23rd 11:59PM
Pass Mark: 65 Points

DATA ANALYTICS — STAGE 3 TASK

Stage 3A: Global Sales & Operations Visual Challenge Stage 3B: Dimensional Data Modelling for Retail Banking Analytics

Requirements / Task Brief

TASK A:

You are a Data Analyst at Northwind Traders, a global gourmet food supplier. Using the provided dataset which covers customers, products, orders, employees, and shippers, build a business intelligence solution that gives the executive team a clear view of global sales performance, shipping efficiency, and salesforce productivity.

The Northwind Traders: DATASET

TASK B:

You've joined the data team at **Palladium Bank**, which operates across Lagos, Abuja, Kano, Port Harcourt, and Ibadan. The bank has 18 months of raw transaction data but no dimensional model. Reports currently run directly on transaction logs making them slow and inconsistent.

The Head of Retail Banking wants a new analytics platform to answer:

- Which customer segments generate the most fee income?
- How does transaction behavior vary by branch and channel?
- Are high-value customers reducing activity?
- Which products drive deposits vs withdrawals?

YOUR TASK: design a dimensional model using the provided DATASET

TASK B: Business Objectives

- Customer Behavior Analysis (by tier, branch, tenure, time)
- Product & Channel Performance (volume, value, trends)
- Branch-Level Reporting (monitor performance & engagement)
- Churn Signals (recency & frequency tracking)
- Time-Based Analysis (MoM, QoQ, YoY comparisons)

Evaluation Criteria / Acceptance Criteria

TASK A

TASK A: Tool Requirement

You may use **any** of the following data visualization tools to complete this task:

- Power BI
- Tableau
- Looker Studio
- Excel

Executive Dashboard

Build an interactive dashboard that answers the five key analytical questions below which is an example and add more business questions. Your dashboard must have **at minimum 3 pages or sections**.

1. Page 1 Sales & Revenue Overview
2. Page 2 Product & Category Intelligence
3. Page 3 Regional, Operational & People Performance

Key Analytical Questions

- Sales Performance: What are the trends in revenue and order volume over time? Are there identifiable seasonal patterns?
- Product Intelligence: Which product categories are the top revenue drivers, and how does the discontinued status of products impact potential sales?
- Regional Insights: Which countries/cities represent the highest market share, and where are shipping (freight) costs disproportionately high?
- Operational Efficiency: How do different shippers compare in terms of cost and speed? Which employees are leading in total sales generated?
- Impact of Strategy: How do discounts affect the overall profit margin across different product lines?

Your dashboard must include at least three working filters (e.g. year, country, category).

TASK B

TASK B: Task Requirements

1. Data Profiling & Dimension Identification

- Classify columns as **dimension, fact, or degenerate dimension**.
- Define each dimension, its attributes, and surrogate key need.
- Decide and justify the **fact table grain** (e.g., per transaction).
- Identify degenerate dimensions and how they'll be handled.

2. Star Schema Design

Create all dimension and fact tables in SQL with:

- Primary & foreign keys
- Appropriate data types
- Surrogate keys
- Comments explaining purpose

3. Slowly Changing Dimensions (SCD)

- Identify which dimensions may change over time.
- Choose and justify SCD strategy (e.g., Type 1 or Type 2) for at least two dimensions.

4. ETL / ELT Strategy

Initial Load:

- Steps to load 18 months of historical data
- Order of table population
- Required transformations

Incremental Loads:

- Detect new records
- Handle SCD updates
- Prevent duplicate facts

Data Quality Checks:

Define at least 4 checks (what, how to detect, and action if failed).

5. Hierarchies, Performance & Scalability

Hierarchies:

Define drill-down structures (e.g., Year → Quarter → Month; State → Branch).

Performance:

Recommend and justify:

- Partitioning strategy
- Indexing approach
- One aggregation table (e.g., monthly revenue by branch)

Submission Format

TASK A:

Deliverables: A google drive folder that contains the following;

1. **Dashboard shareable link (.pbix / .twbx / .xlsx / Looker Studio) which should be in the report.**
2. **Project Report (PDF, minimum 2 pages) link**

NOTE!

Your links should be google drive links. Change default setting on each google drive file to enable anyone to have access to view.

TASK B

Deliverables: A google drive folder that contains the following;

1. **SQL Schema File (.sql) – Full star schema with SCD and date dimension.**
2. **Written Explanation (600 - 800 words) – Justification and design reasoning.**
3. **Schema Diagram – Visual model showing tables and relationships.**

NOTE!

Your links should be google drive links. Change default setting on each google drive file to enable anyone to have access to view.

Submission Link

https://docs.google.com/forms/d/e/1FAIpQLScGWjTI6hiLMxNexmTzsvYx7gAAh4s5bi9k8dtqFnmkdZWQCQ/viewform

Points

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Pass Mark

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